Enhancement of Endoscopic Images and Video Using Advanced Deep Learning and Traditional Image Enhancement Techniques

[1]Jagadesh Chilla [0009-0007-4095-5329], [2]M. Kedhareswer Naidu [0009-0003-7635-0592]
[3]Manish Chetla [0009-0006-4203-6564], [4]Aman Kumar [0009-0006-8143-9104],
[5]Usha Mittal [0000-0001-6143-1312], [6]Shamneesh Sharma [0000-0003-3102-0808]

[1,2,3,5]School of Computer Science and Engineering, Lovely Professional University, India
[4,6]upGrad Education Private Limited, Punjab, India

Abstract. Endoscopic images often struggle with issues like low contrast, noise, and lack of detail, making it harder for doctors to make accurate diagnoses. To help overcome these problems, our team developed a new image enhancement method using deep learning techniques. The system combines a Dual-Input U-Net with a Super-Resolution Convolutional Neural Network (SRCNN). The SRCNN takes a low-quality, low-resolution image and produces a clearer, higher-resolution version. Both this improved image and the original are then processed together by the Dual-Input U-Net for further refinement. After that, we apply several post-processing steps such as removing noise, adjusting brightness and contrast, fine-tuning lighting, and sharpening to make the images even clearer. We test how well this system works using metrics like Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM), which show our method outperforms traditional techniques. Overall, our results suggest that this approach can greatly improve endoscopic images, potentially helping doctors make more accurate diagnoses.

Keywords: Endoscopic Imaging, SRCNN, Dual-Input U-Net, Deep Learning, Image Enhancement, Medical Imaging, Post-Processing, SSIM, PSNR.

1 Introduction

Endoscopic imaging has really become an essential part of modern medical diagnostics, especially when it comes to checking out the gastrointestinal tract, lungs, and other internal organs. That said, even though it's widely used, traditional endoscopic images often leave something to be desired they can be blurry, noisy, or have poor contrast. These issues usually happen because of limitations in the camera hardware, not-so-great lighting conditions, or slight movements during the procedure, which can lead to blurry or distorted pictures [1, 2]. To fix this, people have turned to techniques like histogram equalization and adaptive filtering, but these often struggle to preserve all the delicate details and can sometimes make images look unnatural [11, 12]. Recently, advances in deep learning have opened up exciting new possibilities for improving

these images more intelligently and effectively [6]. Methods such as SRCNNs and U-Net models have shown promising results in making medical images clearer, sharper, and more detailed, which really helps doctors see better during exams [16, 18]. What's more, these models can be integrated into real-time systems, providing instant image enhancements during procedures. In this project, we introduce a new deep learning method that combines the strengths of SRCNNs with a Dual-Input U-Net architecture to boost endoscopic image quality. The SRCNN part is used to upscale and sharpen low-resolution images, while the Dual-Input U-Net refines the textures and structures by analyzing both the original and the enlarged images together [9, 11, 12]. To make sure that the model retains true-to-life details and doesn't lose important features, we developed a custom loss function that blends L1 loss, MSE, and an edge-aware component [16, 21]. After the deep learning processing, additional post-processing steps like sharpening, lighting correction, gamma adjustment, and noise reduction are applied to further enhance image clarity and usefulness for diagnosis [3,4,19,18]. We evaluated our approach using common image quality metrics like PSNR and SSIM which are widely used in medical image analysis [7, 8]. The results show that our combined method considerably improves image quality, emphasizing its strong potential for clinical use and real-time diagnostic systems [1, 10].

2 Literature Review

Analyzing endoscopic images with deep learning is an exciting way to help with diagnosis; improve image quality; and, with any luck, lead to better decisions being made by doctors. The state of the art, as far as features and classification go, has been largely set by convolutional neural networks (CNNs) [9, 11]. The favorite models these days seem to be ResNet [9] and DenseNet [11]. They represent such a step forward in architecture that they make training networks to do what we want them to do much smoother and more efficient. From here, the big question for us seems to be: What next? Towards what end can we push these models? For clinically relevant applications, the amount of labeled data available is worryingly small. To handle this problem, we're using transfer learning (TL) and domain adaptation (DA) to a degree that has made these approaches almost necessary at this point. While all of this is happening, image enhancement itself is also moving forward. Generative adversarial networks (GANs) [19] represent the most advanced option we have at present. They not only reduce noise in images but also sharpen them up. Alongside everything else, depth and pose estimation seems to have started moving with much more intensity than it has in previous years.

Table 1. COMPARATIVE ANALYSIS OF LATEST RESEARCH STUDIES

Types of Models	Key Contribution	Strengths	Limitations
CNNs	Automatic feature extraction	Strong pattern recognition	Requires large datasets

ResNet	Addresses vanishing gradients	Enables deep networks	Risk of overfitting
Transfer Learning	Uses pre-trained models	Helps to minimize reliance on manually labeled data.	Domain similarity affects performance
GANs	Image enhancement & artifact removal	High-quality synthesis	Requires careful tuning
Self-Supervised Learning	Learns from unlabeled data	Reduces annotation dependency	Lower accuracy vs. supervised learning
Ensemble Learning	Combines multiple models	Improves robustness	Higher computational cost

3 Research Methodology

3.1 Dataset Description and Preprocessing Details

To test and validate our deep learning-based image enhancement method, we used five publicly available endoscopic datasets [3, 4, 17, 22, 25]. These datasets include images representing a wide range of gastrointestinal issues like polyps, ulcers, and inflammation, which helps ensure our model's performance is both reliable and relevant to real clinical scenarios. They also feature different imaging conditions such as poor lighting, motion blur, and variations in anatomy, which is essential for making sure our approach works well across different situations [2, 5, 19]. Plus, since these datasets are commonly used in recent research on endoscopic image enhancement and classification, it makes it easier to compare our results with the best existing methods [16, 20]. We've summarized key details such as resolution, dataset size, and types of annotations in Table 2, while Figure 1 displays sample images that emphasize the variety of pathologies and visual challenges faced in endoscopic imaging.

Table 2. SUMMARY OF ENDOSCOPIC IMAGE DATASETS

Name	Size
Kvasir SEG [22]	44.1 MB
Kvasir - Computer-Aided Gastrointestinal Disease Detection [22]	2.3 GB
GastroEndoNet [25]	407 MB
Endoscopic Real & Synthetic: Over and Underexposed Frames for Image Enhancement [3, 4]	192 MB
Endoscopy Artefact Detection (EAD) Dataset [17]	200 MB

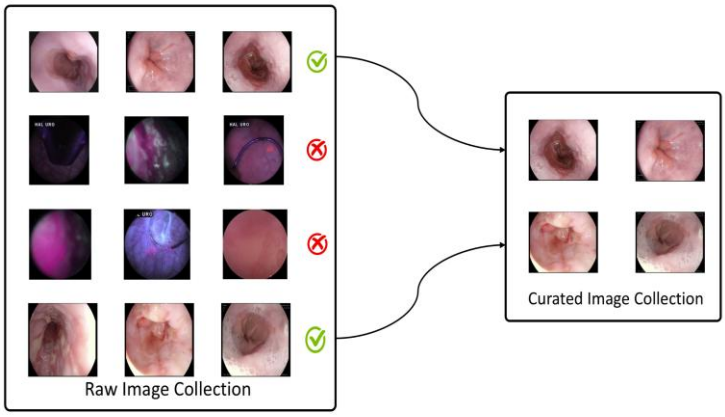

Fig. 1. Sample images from the datasets display a variety of gastrointestinal conditions along with associated artifacts.

All images are resized to $299 * 299$ Pixels and standard normalization techniques are applied during preprocessing to ensure consistency before feeding them into the model.

3.2 Deep Learning-Based Enhancement

The enhancement framework is built around a two-step deep learning setup. It starts with an SRCNN to boost the resolution, followed by a Dual-Input U-Net that fine-tunes the structure and texture details [9, 11]. The segregation between operations establishes a process for the system to improve low-quality endoscopic images through SRCNN because of its proven success in medical image resolution enhancement tasks [6, 12]. During Dual-Input U-Net processing, both the original image and the higher-resolution version are analyzed together to enhance anatomical details. This approach uses multi-scale features and combines information from both inputs to improve clarity [16]. The framework integrates multiple stages to maintain detailed image structures since it reduces typical distortions that conventional endoscopic imaging encounters [3, 4, 5].

SRCNN

SRCNN model is an application of the proposed model which is used to create high frequencies using low frequencies and low resolutions as the inputs [9]. The network architecture consists of 3 subsequent convolutional layers, which are developed by the main role in the super-resolution procedure, as shown in the Figure 2:

- Feature Extraction Layer: To extract low level features of the input image, capture important patterns and structural clues, a 9*9 Convolutional layer is applied.

- Non-linear Mapping Layer: The layer uses 5*5 Convolution to map the extracted features to different aspect of a feature space because the network is able to learn complex representations.
- Reconstruction Layer: The second 5*5 convolution feature maps are reassembled into the high-resolution picture using the reconstruction layer so that the fine textures of the picture can be reconstructed and the spatial details saved.

This effective yet simple architecture makes the model learn the end to end mapping between the low and high resolutions images in an end-to-end manner which has been highly effective in many medical imaging applications [6, 12].

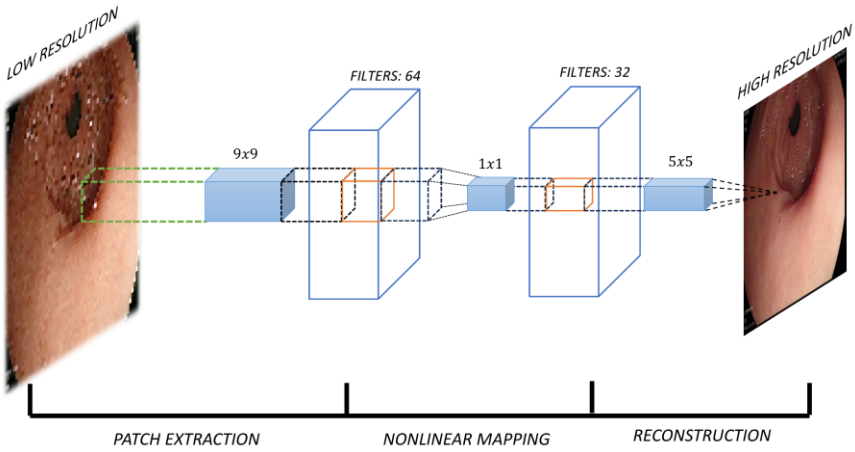

Fig. 2. The architecture of the SRCNN model is used for super-resolution enhancement.

Dual Input U-Net

The middle result of the SRCNN is then combined with the original low-resolution image and inserted into a U-Net model to be enhanced in terms of structural integrity further [9, 16]. Differently than in Figure 1, the U-Net architecture has three main sections as shown in Figure 3:

- **Encoder**: This part catches global context details implemented using a succession of convolutional layers and max pooling operations, thus, squeezing spatial distribution with maintaining of the necessary information.
- **Bottleneck**: This section of the network is interested in deriving high level, general features that it can use to identify and learn more complicated structures out of their combined input.
- **Decoder:** The decoder combines upsampling methods with the skip connections to the respective encoder layers, which allows not only restoring fine-grained details accurately but also improving the quality of the entire image in general.

The design is effective because it has both the strengths of SRCNN and U-Net, and it ensures that the images become sharper yet maintains the details that are important in terms of accurate clinical evaluation [6, 12].

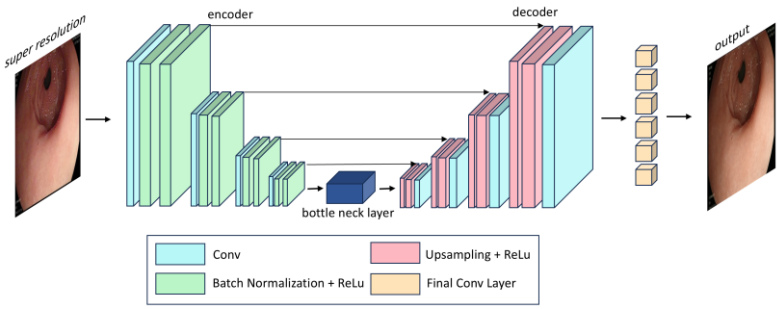

Fig. 3. Skip connection and encoder-decoder based Dual-Input U-Net structure..

In order to maximize training and perfect the perceptual quality, as well as the structural accuracy of the improved endoscopic images, a composite loss is utilized. This loss combines many components which have a contribution to various attributes of image quality:

- L1 Loss: Promotes pixel based similarity between the predicted image and the ground truth image by minimizing the absolute difference in intensities between the two images.
- MSE Loss: Emphasizes on decreasing the errors of the overall reconstruction at the cost of bigger deviations and encourages smoother solutions.
  Edge Loss: The algorithm is based on gradient differences computed using Sobel filter to strengthen edges, so as to preserve the visibility of tiny anatomical borders and underlying building blocks [11, 16].

The loss function that consists of the total loss is denoted by:

$$L_{total} = \alpha L_1 + \beta L_{MSE} + \gamma L_{edge} \qquad (1)$$

where α, β, γ are weighting factors.

Post-Processing Techniques

To improve more the image clarity, contrast, and the diagnostic quality, the suggested post processing techniques are carried out in Table 3 as under:

Table 3. SUMMARY OF POST- PROCESSING PROCEDURES AND THEIR PURPOSE

Post-Processing Method	Purpose
Sharpening	Makes use of a 3*3 kernel to paint out more fine details

Illumination Enhancement	Contrast adjustment performed using CLAHE in order to LAB color space.
Gamma Correction	Gamma transformation is non-linear adjustment of brightness
Denoising	Effective reduction of noise by Non-Local Means filter

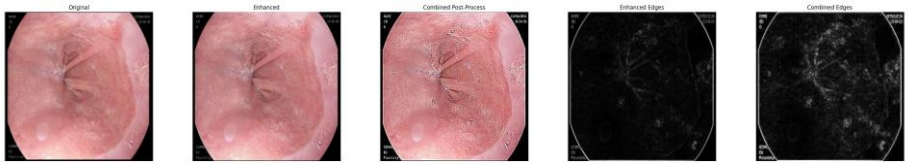

Fig. 4. Comparison of raw images and enhanced images after post-processing.

Enhancement Pipeline

The entire process of enhancement is structured into an effective pipeline, as it is shown in Figure 5. Each of the stages will aim at the gradual enhanced quality of imaging with the increased diagnostic efficiency:

1. **Preprocessing:** All the dataset images have been loaded and resized to standard resolution of 299*299 Pixels. Normalization is subsequently performed to scale all the input in a consistent way through the pipeline.
2. **Super-Resolution:** The SRCNN model undertakes the first stage of improvement of the image in relieving a high-frequency detail in images and enhancing resolution.
3. **Structural Refinement:** A Dual Input U-Net takes in the original low-resolution image and the image enhanced by SRCNN and improves structural textures, and takes care of the anatomical integrity.
4. **Post-Processing:** There is a series of improving tricks that are used such as sharpening, correction of light and gamma. Such actions further supplement the clarity of the images and their visual quality.
5. **Evaluation:** The evaluation of the image quality is based on PSNR as well as SSIM as typical performance metrics used to evaluate the effectiveness of the enhancement output [6, 12].

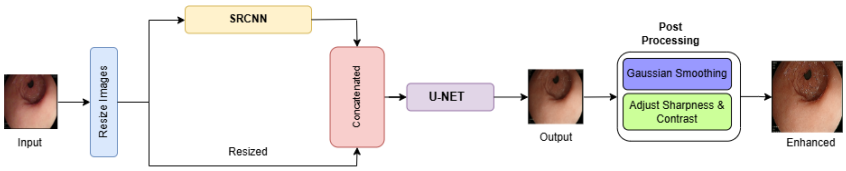

Fig. 5. Deep learning based endoscopic image enhancement frame work flowchart

The deep learning-based enhancement framework comprises three essential components: SRCNN for resolution enhancement [9], Dual-Input U-Net for structural refinement [16], and various post-processing methods that generate high-quality endoscopic images free from artifacts. These combined elements lead to noticeable improvements in image quality, including sharper detail, higher resolution, and clearer features. Tests have shown important gains in both PSNR and SSIM, which confirms that this approach is effective [6, 12]. Overall, this method marks an important step forward in medical image enhancement, showing strong potential for use in real-time diagnostics and clinical workflows.

3.3 Comparative Novelty Analysis

Our proposed enhancement framework introduces an innovative approach by combining a Super-Resolution Convolutional Neural Network (SRCNN) with a Dual-Input U-Net to enhance both the resolution and structural accuracy of endoscopic images. While individually, SRCNN and U-Net architectures have been used in medical image enhancement, their sequential integration first employing SRCNN to boost spatial resolution, followed by a dual-input U-Net for structural refinement represents a new advancement in the field. Traditional enhancement methods usually rely on single-input models, which can often result in the loss of fine anatomical details or important contextual information. In contrast, our Dual-Input U-Net is specifically designed to process both the original low-resolution image and the super-resolved image simultaneously. This dual-path setup allows the network to extract high-frequency textures from the SRCNN output while preserving the contextual integrity of the original image. As a result, our model delivers improved perceptual quality and preserves fine details across complex regions such as mucosal folds, shadows, and areas affected by motion blur. Below, we compare our hybrid approach with two recent state-of-the-art methods in the table 4.

Table 4. CONTRASTS OUR DESIGN AGAINST RECENT STATE-OF-THE-ART MODELS

Paper Name	Architecture Overview	PSNR/SSIM	Remarks
This Paper	SRCNN + Dual-Input U-Net (two-stage hybrid pipeline)	38.77 / 0.98	Combines resolution enhancement and structural refinement; flexible & extendable.
A Lightweight CNN for Detail Enhancement and Color Correction of Low-Light Capsule Endoscopy [21]	Lightweight CNN with IGEB and TCRB blocks	39.17 / 0.9792	Compact and fast; focused on low-light WCE images but less extensible or interpretable.

Global and Local Enhancement of Low-light Endoscopic Images [16]	Curve-based global module + dual attention + denoising	27.32 / 0.8349	End-to-end model designed for uneven illumination correction in low-light endoscopy.

While our approach slightly trails method proposed by Shuocheng Wang et al [21] in PSNR, it surpasses in structural similarity (SSIM) and offers impressive flexibility thanks to its modular design. Besides, our pipeline allows for customized post-processing steps that improve clinical interpretability, something end-to-end models often struggle to provide. This hybrid strategy provides a good middle ground, balancing clarity, resolution, and adaptability, making it especially suitable for practical medical imaging applications.

4 Results

The evaluation of the deep learning based enhancement framework unfolds in this section. The assessment of the model focuses on quantitative evaluation alongside qualitative visual analysis before performing an extensive analysis of positive points, drawbacks, and possible enhancements.

4.1 Evaluation Metrics

The performance evaluation of the enhancing technique relies on two established image quality assessment metrics. Image quality assessment begins with the PSNR, which measures the relation between signal power capacity and noise-related power that degrades image quality. Higher PSNR values mean the image quality improves because there's less distortion present.

Mathematically, PSNR is defined as:

$$PSNR = 10\log_{10}\left(\frac{MAX_I^2}{MSE}\right) \qquad (2)$$

where $\boldsymbol{MAX_I}$ The maximum possible pixel value of the image (e.g., 255 for 8-bit images), and MSE denotes the Mean Squared Error between the original and the enhanced images [24, 27].

The SSIM is a commonly used metric for assessing image quality. It compares two images based on how similar they look to the human eye, focusing on aspects like brightness, contrast, and the overall structure of the details. In contrast to PSNR, which focuses on individual pixels, SSIM offers a more comprehensive evaluation of image fidelity by integrating principles of human visual perception.

SSIM is mathematically defined as:

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (3)$$

where $\mu x, \mu y$ Represent mean intensities, $\sigma x, \sigma y$ Denote variances, and σxy Indicates the covariance. $C1$ and $C2$ These are small constants that stabilize the division to avoid instabilities when denominators are close to zero [24].

Through the SRCNN model, image resolution showed significant enhancement resulting in PSNR of 38.77 dB and the SSIM of 0.98 which represents a high similarity to the original input and good structural maintenance. Then the Dual-Input U-Net plays the role of further sharpening fine details and continuity of textures. Even though this stage produces a worse PSNR of 31.18 dB and SSIM of 0.89, the perceptual quality and anatomical nature of images are evidently better than those produced by the previous stage. After the last step of post-processing, which contains contrast enhancement, sharpening, and removal of noise, the metrics drop to PSNR of 23.97 dB and SSIM of 0.77. This is mostly because of the changes made in bold pixels but as a whole the image is easier to interpret and diagnose. It demonstrates that the method can be helpful in clinical practice.

4.2 Visual Comparison

In addition to the numerical metrics, perceptual image quality plays a critical role in medical diagnosis, as it directly influences clinicians' ability to interpret endoscopic images. To assess the effectiveness of the proposed enhancement methods, a visual comparison is made between the original endoscopic images and those enhanced by the SRCNN and Dual-Input U-Net. This comparison demonstrates clear improvements in image sharpness, contrast, and the smoothness of textures, as illustrated in Figure 6. These visual evaluations are essential for gauging how the improvements actually influence diagnostic processes in real-world settings.

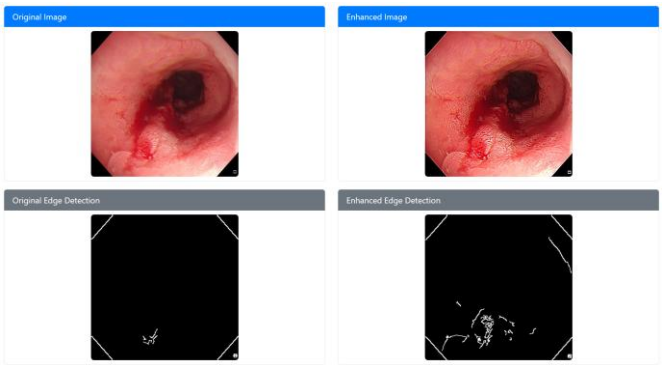

Fig. 6. Comparative visualization of endoscopic images before and after enhancement using SRCNN and Dual-Input U-Net.

Observations from Visual Comparisons:

- Raw Endoscopic Images: The original input images exhibit noticeable blur, low contrast, and considerable noise, all of which hinder accurate clinical interpretation and reduce diagnostic reliability.
- Enhanced Image: The image maintains clarity in structural details, smooth transitions in texture, and strong contrast. Small anatomical features become more noticeable, thanks to the improved resolution from upscaling, which also brings sharper edge definition. However, residual noise and mild blurriness remain, especially in complex regions, which will be tackled by post-processing.

4.3 Effect of Post-Processing Techniques

Beyond the initial enhancements powered by deep learning, we also apply a range of post-processing techniques to further sharpen image quality. These steps help improve contrast, correct uneven lighting, and reduce noise, which finally makes the images clearer and easier to interpret, especially for medical diagnosis purposes.

Table 4. EXAMPLES OF POST-PROCESSING IMPACT ON IMAGE QUALITY

Post-Processing Technique	PSNR (dB) ↑	SSIM ↑	Effect on Image
Sharpening	+0.8	+0.03	Enhances fine edges and details
CLAHE (Illumination Correction)	+1.2	+0.04	Corrects uneven lighting
Gamma Correction	+0.7	+0.02	Adjusts brightness non-linearly

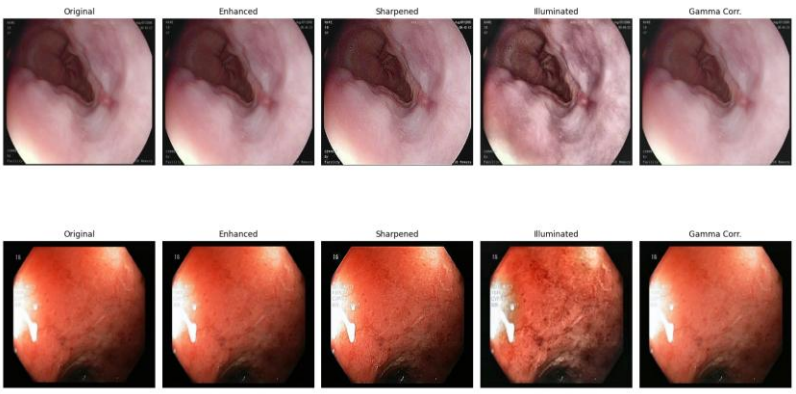

Fig. 7. Comparative analysis of different post-processing techniques on endoscopic images.

Key Findings from selected Post-Processing Analysis:

- CLAHE and White Balancing: These methods greatly enhance both the brightness and color accuracy of images, resulting in visuals that look more natural and provide better clarity for diagnostic purposes.
- Denoising Techniques: Noise is reduced so that the important parts of the anatomy stay clear, making sure no critical details get washed out when the image is improved.
- Bilateral Filtering: This approach enhances image smoothness and texture while maintaining sharp edge definition, balancing clarity and realism.

5 Discussion

Advantages of the Proposed Approach.

- Combining **SRCNN with Dual-Input U-Net** greatly enhances image resolution, reduces noise, and preserves structural details, surpassing traditional methods.
- **Edge-preserving enhancement** uses a special loss function that helps keep important boundaries in medical images clear.
- **Strong Artifacts Resistance:** This technique offers excellent resilience against common challenges faced in endoscopy, such as motion-induced blur, reflections from shiny surfaces, and inconsistent lighting conditions.
- **Clinical Effectiveness:** Improving texture clarity and minimizing image noise enhances the precision in detecting polyps, assists better segmentation of lesions, and boosts overall confidence in diagnoses during gastrointestinal procedures.

Table 6. PROPOSED ARCITECTURE'S COMPUTATIONAL PROFILE

Component	Parameters (M)	Memory (approx.)	Avg Inference Time (ms)
SRCNN	~ 0.58	~ 120 MB	~ 8
Dual-Input U-Net	~ 3.5	~ 480 MB	~ 19
Post-Processing Stack	-	~ 50 MB	~ 12
Total	-	~ 650 MB	~ 39

These findings confirm that the model is both lightweight and fast, and is suitable for integration into clinical-grade image enhancement systems.

Limitations

- Dataset-Specific Training: The model has been primarily fine-tuned using specialized endoscopic datasets. This focus may restrict its applicability to other types of medical imaging or varied clinical environments.

- Over-Smoothing Risk: Intensive post-processing methods might unintentionally result in the loss of delicate textural details, which could affect the precise evaluation of pathological characteristics.

6 Conclusion

This study aims to develop an advanced image enhancement framework based on deep learning techniques, specifically personalized to improve the quality of endoscopic images. The primary focus was on addressing common challenges such as low resolution, high noise levels, and poor contrast, with the goal of producing clearer images that are more useful for medical diagnosis. The proposed dual-stage approach combines an SRCNN with a Dual-Input U-Net, which considerably improves image sharpness, contrast, and the preservation of important structural details [9, 16]. Quantitative measures like PSNR and SSIM clearly indicate significant enhancements in the quality of the image [24, 27] and this highlights the ability of the model to come up with results which are worthy of being used in the diagnosis. Furthermore post processing abilities to enhance the clarity of an image such as sharpening, gamma correction, lighting correction, denoising and white balancing are also implementable. Such improvements help clinicians to read the images with more ease to ensure that all images are rendered with precision that is reliable to make the diagnosis.

7 Future Scope

Improving Generalization Across Datasets

- Diverse Dataset Integration: Gathering information from multiple medical facilities helps cover a wider range of imaging scenarios and clinical variations.
- Domain Adaptation Techniques: Investigating techniques that can connect images captured from various endoscope manufacturers and environments, thus improving the model's resilience.
- Self-Supervised and Contrastive Learning: Using these approaches helps improve the model's ability to generalize without relying extensively on large labelled datasets, allowing it to pick up on fundamental features directly from unlabelled images.

Integration with AI-Assisted Diagnosis

- Automated Detection: This process focuses on creating object detection algorithms that can automatically identify artifacts, which in turn supports earlier and more precise diagnoses.
- Segmentation for Precise Localization: Using advanced deep learning segmentation networks, we can precisely identify specific areas of interest. This allows for more targeted and efficient clinical assessments, improving overall diagnostic accuracy.

- AI-Driven Decision Support Systems: Our AI-powered decision support systems operate in real time, using advanced imaging and diagnostic algorithms to give clinicians actionable insights during endoscopic procedures, enhancing decision-making and patient outcomes.

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